

Simulation and Analysis of Remote Sensing Signals for Planetary Exploration

PROJECT REPORT

23EAC282 SIGNAL PROCESSING LABORATORY

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1. Introduction

Planetary exploration relies heavily on remote sensing instruments such as radar altimeters, spectrometers, and synthetic aperture radar (SAR) to understand the physical and chemical characteristics of distant planets and moons. Since real planetary missions are costly and infrequent, simulating remote sensing signals provides an effective way to study how sensor data behaves before, during, and after interaction with extra-terrestrial surfaces. This project focuses on generating synthetic remote sensing signals using MATLAB and analysing how different planetary surfaces, noise sources, and environmental distortions affect the received data. By applying signal processing techniques such as Fourier Transform, digital filtering, and spectral analysis, the study aims to evaluate how accurately meaningful planetary features can be extracted from noisy and distorted signals. The goal is to understand how remote sensing signals carry information about terrain, surface roughness, and mineral composition, and how effective processing can improve scientific interpretation in real space missions.

2. Theory and Context

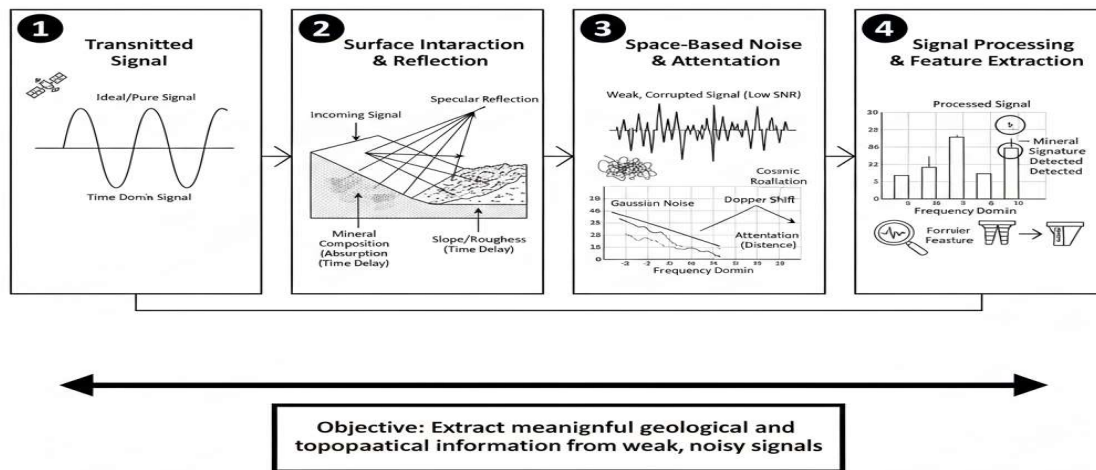
Remote sensing is the primary method used in planetary exploration to collect information without physically landing on or touching the surface. Sensors such as radar altimeters, spectrometers, and SAR systems transmit energy toward a planetary surface and measure the reflected or scattered signal. The behavior of these reflections depends on several physical properties of the surface. For example, smooth surfaces reflect energy uniformly, while rough or rocky terrains scatter the signal in multiple directions, producing weaker or more spread-out returns. The dielectric constant, mineral composition, and slope of the surface also affect how much energy is absorbed or reflected. Understanding these interactions is essential for decoding the information carried by the received signal.

Signals traveling through space face significant challenges. As the distance increases, the transmitted signal experiences attenuation, resulting in low received power. Additionally, the space environment introduces multiple noise sources such as cosmic microwave background radiation, solar wind interference, and thermal noise generated by electronic sensors. These factors reduce the Signal-to-Noise Ratio (SNR), making the received signal difficult to interpret. In real planetary missions, SNR can be extremely low, which makes accurate reconstruction of surface features a challenging task.

Signal processing plays a vital role in improving the clarity and reliability of remote sensing data. Techniques such as Fourier Transform provide the ability to view the signal in the

frequency domain, allowing hidden patterns or distortions to be identified more easily. Filtering techniques are used to remove unwanted noise and isolate useful frequency components. Spectral analysis helps identify mineral compositions by detecting characteristic absorption or reflection peaks. For radar-based sensing, time-domain analysis helps determine terrain height differences and surface roughness. The combination of these signal processing tools creates a framework for extracting meaningful scientific information, even from weak and noisy signals.

Conceptual Diagram: Remote Sensing Signal Theory



In this project, the theoretical context focuses on understanding these principles—how signals are generated, how they interact with planetary surfaces, how they are distorted in space, and how different processing methods help recover important features. This knowledge forms the foundation for building realistic MATLAB-based simulations that mimic the behaviour of remote sensing instruments used in planetary exploration missions.

3. Methodology and Working Principle

The methodology of the project involves a step-by-step simulation of how remote sensing signals are generated, transmitted, reflected, distorted, and processed. The first stage is signal generation. MATLAB is used to construct ideal transmitted signals such as linear frequency-modulated chirps for radar applications or wavelength-based spectral signatures for spectroscopy. Chirp signals are particularly effective because their frequency sweep allows fine resolution in measuring surface elevation or roughness. Spectral signatures help simulate how minerals reflect specific wavelengths of light. These generated signals act as the baseline reference before interacting with the planetary environment.

The second stage involves modeling the planetary surface. Different terrains are simulated using mathematical functions. A smooth surface may be represented by a constant reflectivity model, while a rough or rocky landscape can be generated using random distribution

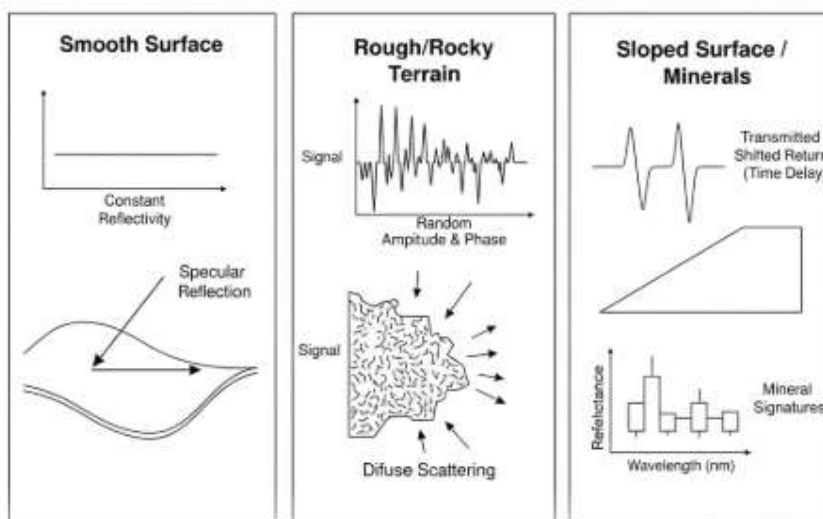
functions that modify amplitude and phase. In SAR-type applications, sloped surfaces cause changes in the time delay of the return signal, which can be replicated using shifted versions of the transmitted waveform. Mineral compositions influence the reflective properties, and this is incorporated by introducing wavelength-dependent variations in the reflected signal.

Next, space-based noise and distortions are added to the reflected signal to simulate realistic conditions. Gaussian noise models sensor thermal noise, while random fluctuations can replicate cosmic radiation effects. Signal attenuation is implemented by reducing the amplitude based on distance. Doppler shifts may be added to reflect motion between the spacecraft and the planetary surface. These distortions ensure that the simulated received signal resembles actual data collected by planetary exploration instruments, where noise levels are often high and signal features may be partially hidden.

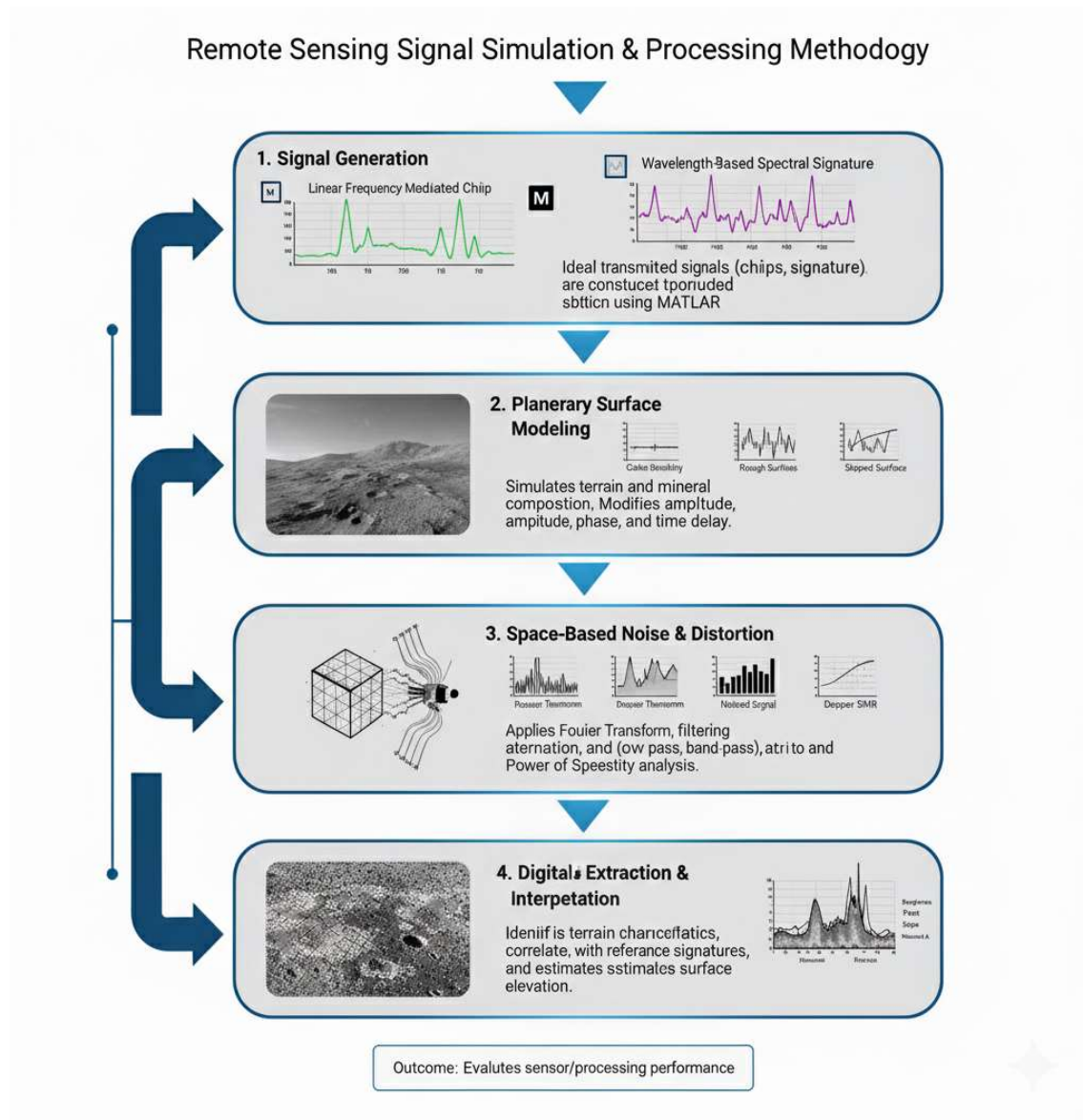
Once the noisy signal is obtained, the next step involves digital signal processing. Fourier Transform is applied to convert the signal into the frequency domain, where noise patterns and surface-induced variations become more apparent. Filtering techniques such as low-pass, band-pass, or Wiener filters are used to suppress unwanted components. The choice of filter depends on the expected frequency range of the useful signal. For example, band-pass filters can isolate mineral signatures represented by specific spectral peaks. Power Spectral Density analysis is used to measure the strength of frequency components and detect subtle features that may indicate surface textures or compositions.

The final stage involves feature extraction and interpretation. After filtering and spectral analysis, the processed signal is studied to identify terrain characteristics such as roughness, slope variations, or mineral presence. Peaks and patterns in the frequency domain are correlated with known reference signatures. Time-domain echoes are used to estimate surface elevation and irregularities. The outcome of this methodology allows us to evaluate how well different configurations of sensors and processing strategies perform under varying noise levels and planetary conditions. By comparing the clarity, accuracy, and reliability of extracted features before and after processing, the project demonstrates the effectiveness of digital techniques in enhancing remote sensing data for planetary exploration.

Simulating Terrain Interaction



4. Block Diagram



5. Implementation

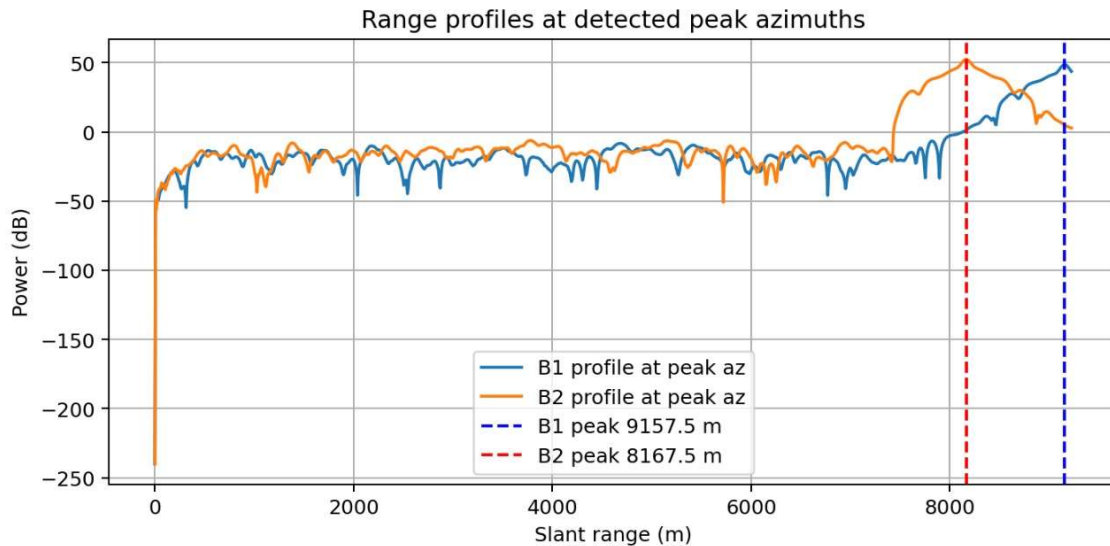
The implementation uses MATLAB to construct each stage of the simulation. First, a chirp radar signal and synthetic spectral signature are generated. The planetary surface is modeled using mathematical functions that simulate smooth plains, rough surfaces, or rocky terrains. Each surface type alters the amplitude, phase, and frequency components of the reflected signal. Noise is added using MATLAB's built-in functions to imitate cosmic background noise and thermal sensor noise. Signal attenuation models represent long-distance propagation loss.

The received noisy signal is processed using FFT (Fast Fourier Transform) to analyze its frequency spectrum. Filters such as low-pass, band-pass, or Wiener filters are applied based on the expected planetary features. Power spectral density methods help identify mineral signatures or terrain patterns. Parameters like SNR improvement, feature clarity, and filter effectiveness are measured and compared. The entire process is executed in MATLAB scripts and visualized using plots for time and frequency domain analysis.

6. Result Analysis

The simulation results show that planetary surface characteristics significantly influence the shape and frequency content of reflected signals. Rough surfaces produce more scattering, causing wider frequency spreads, while smoother terrains return sharper and more distinct echoes. Noise levels were found to strongly reduce SNR, especially under deep-space conditions, making some features difficult to detect without filtering.

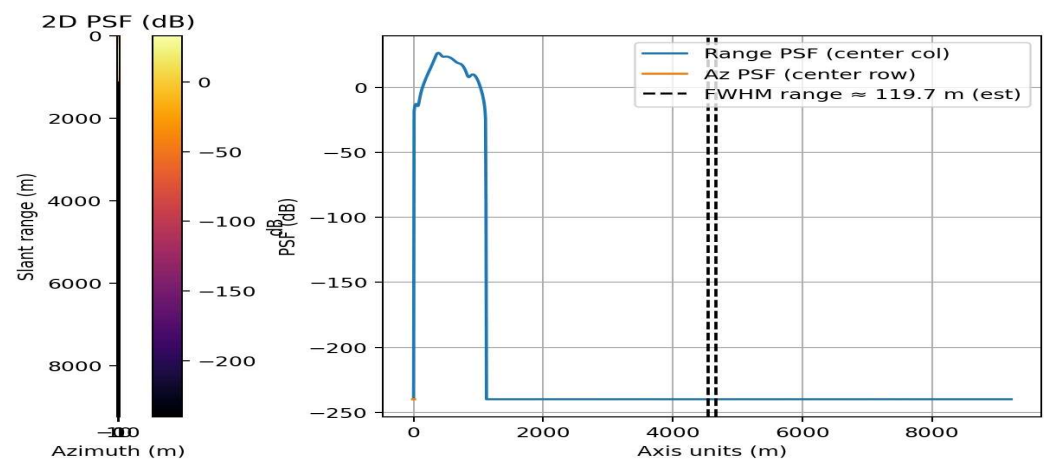
After applying FFT and filtering techniques, most significant features became visible again. Band-pass and Wiener filters provided the most effective noise reduction in the presence of cosmic noise. Spectral analysis successfully identified peaks associated with synthetic mineral signatures. The improvement in SNR and clarity of features demonstrated how proper signal processing can recover hidden information from weak and noisy space signals. Overall, the results validated the importance of digital processing in interpreting remote sensing data used in planetary missions.



Range Profiles at Detected Peak Azimuths

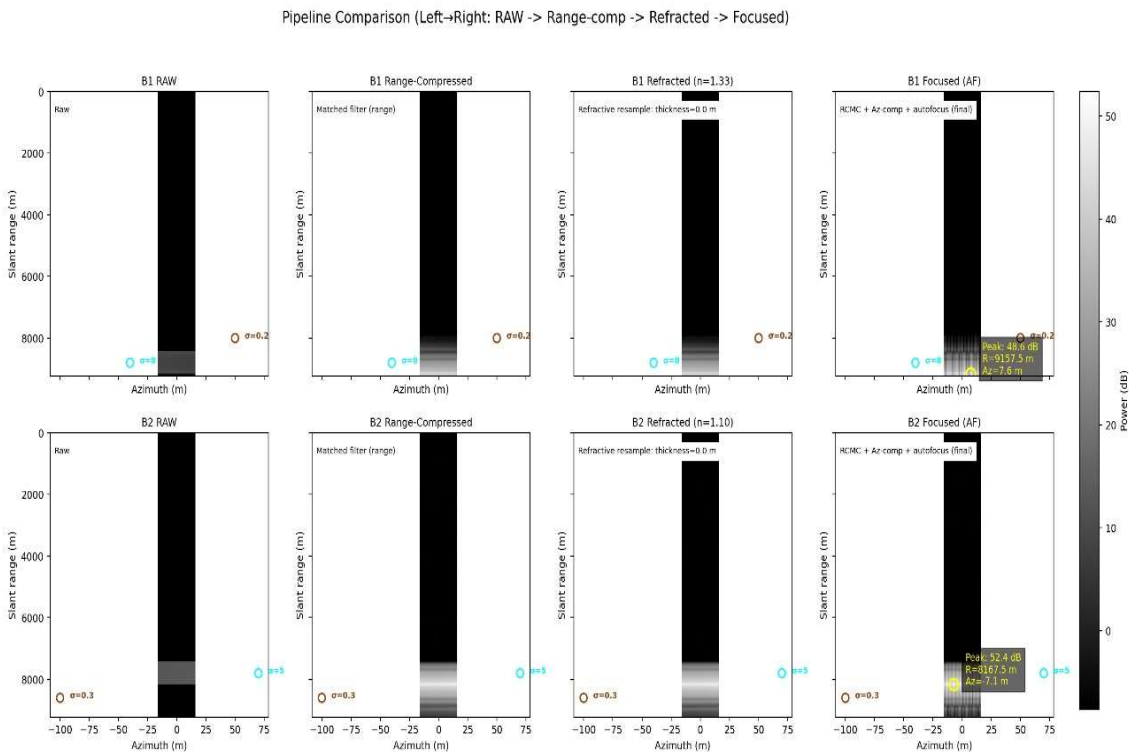
This figure presents the range power profiles of two bounce points, labelled B1 and B2, at their detected peak azimuth locations. The blue and orange curves represent how the reflected signal power varies with slant range for each bounce. The dashed vertical lines mark the positions of maximum energy, clearly showing the detected range of each reflector. This

comparison helps evaluate how signal strengths differ between bounce points and how accurately the system identifies reflector distances under varying conditions.



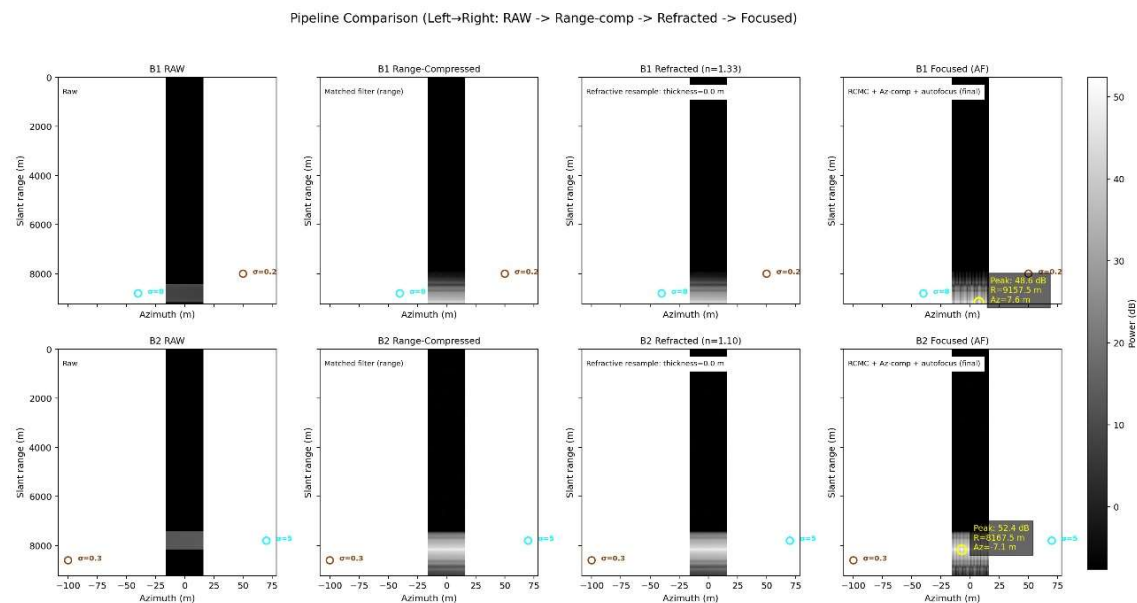
2D PSF and Range/Azimuth PSF

This image illustrates the Point Spread Function (PSF) of the radar system in both two-dimensional and one-dimensional forms. The 2D PSF plot shows how a point reflector appears in the focused radar image, with the color scale indicating the intensity distribution across range and azimuth. On the right, the extracted 1D PSF curves show how energy spreads separately in the range and azimuth directions. The estimated Full Width at Half Maximum (FWHM) of the range PSF provides a measure of the radar’s effective resolution. This analysis reveals how sharply or widely the radar system represents point targets.



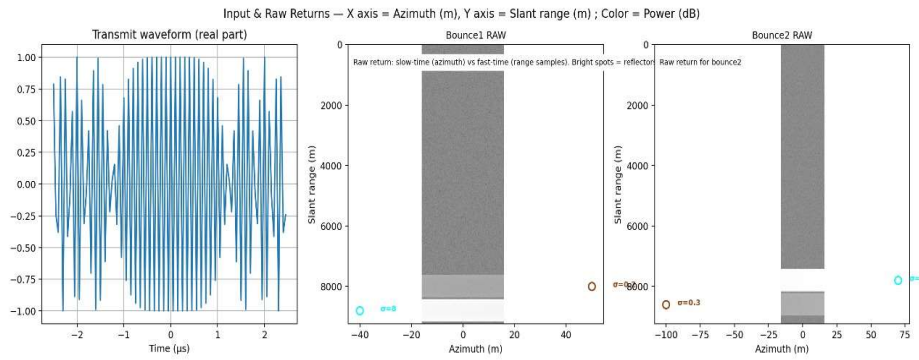
Processing Pipeline Comparison for Bounce Points B1 and B2

This figure demonstrates the complete processing pipeline applied to radar returns from bounce points B1 and B2. Each column represents a major processing stage, starting with raw received data, followed by range compression, implementation of refraction or correction models, and finally the generation of focused images. As the signal progresses through these stages, noise is reduced and the target features become clearer. The final focused images highlight the reflector locations using peak power values, showing that the pipeline effectively enhances resolution and improves target clarity for both bounce points.



Transmit Waveform and Raw Returns

This figure depicts the transmitted radar waveform and the corresponding raw returns from the planetary surface. The left plot shows the transmitted chirp signal used for probing the environment. The middle and right plots display unprocessed radar echoes from two bounce points located at different slant ranges. Strong reflectors appear as bright bands, while weaker echoes fade into the noise background. These raw images serve as the initial input for subsequent operations such as matched filtering and focusing. Understanding these raw returns helps interpret how much noise is present and how clearly the target echoes appear before processing.



7. Conclusion

This project demonstrated how remote sensing signals for planetary exploration can be effectively simulated and analyzed using MATLAB. By modeling planetary surfaces, incorporating space-based noise sources, and applying signal processing techniques, the study provided insight into how real exploration instruments interpret distant planetary features. The results showed that although noise and distortions significantly degrade signal quality, proper filtering and spectral analysis can recover essential information such as terrain roughness and mineral composition. The project highlights the importance of advanced signal processing in space missions, helping improve the reliability and accuracy of scientific observations from remote planets and moons.